

Problem Set 12:

Submission deadline: 12:00, Friday, July 10, 2026; Solutions: Friday, July 17, 2026

Problem 1 Doppler Effect

An observer O sees a light ray that moves in the $x - y$ plane at an angle θ with respect to the x -axis, i.e. $k^\mu = (k, k \cos\theta, k \sin\theta, 0)$ with $k = \omega/c$. Another observer O' moves with respect to O with velocity v in the x -direction

(1.a) Show that the frequency ω' observed by O' is given by

$$\omega' = \gamma \left(1 - \frac{v}{c} \cos\theta \right) \omega. \quad (1)$$

This frequency change is called the *Doppler shift*.

(1.b) The observer O' sees the light ray move at an angle θ' . Show that

$$\tan\theta' = \frac{\sin\theta}{\gamma \left(\cos\theta - \frac{v}{c} \right)}. \quad (2)$$

This change of angle is called *aberration*.

Now replace the observer O' by a perfect mirror perpendicular to the x -axis, which is still moving with velocity v in the x -direction. In addition, for simplicity we choose $\theta = 0$, so that the light ray hits the mirror perpendicularly.

(1.c) Determine the frequency ω_r of the reflected wave observed by O .

Solution:

(1.a) The Lorentz transformation Γ^μ_ν , which transforms from the reference frame of the observer O to the reference frame of the observer O' , is given by

$$\Gamma^\mu_\nu = \begin{pmatrix} \gamma & -\gamma \frac{v}{c} & 0 & 0 \\ -\gamma \frac{v}{c} & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (3)$$

Thus we obtain

$$k^\mu \rightarrow k'^\mu = \Gamma^\mu_\nu k^\nu = \begin{pmatrix} \gamma k - \gamma \frac{v}{c} k \cos\theta \\ -\gamma \frac{v}{c} k + \gamma k \cos\theta \\ k \sin\theta \\ 0 \end{pmatrix}. \quad (4)$$

From this one reads off that

$$\omega' = ck'^0 = c\gamma k \left(1 - \frac{v}{c} \cos\theta \right) = \gamma \left(1 - \frac{v}{c} \cos\theta \right) \omega. \quad (5)$$

(1.b) Here we can also simply read off from Eq. (18)

$$\tan\theta' = \frac{k'^2}{k'^1} = \frac{\sin\theta}{\gamma \left(\cos\theta - \frac{v}{c} \right)}. \quad (6)$$

(1.c) In the reference frame O' of the mirror we find

$$\omega'_r = \omega', \quad (7)$$

$$\mathbf{k}'_r = -\mathbf{k}'. \quad (8)$$

In four-vector notation, we therefore have

$$k'^\mu_r = (k', -k', 0, 0) = \left(\gamma \left(1 - \frac{v}{c} \right) k, -\gamma \left(1 - \frac{v}{c} \right) k, 0, 0 \right). \quad (9)$$

Transforming this back into the reference frame of the observer O , we obtain

$$(k_r)^\mu = (\Gamma^{-1})^\mu{}_\nu (k'_r)^\nu = \begin{pmatrix} \gamma k'_r + \gamma \frac{v}{c} k'_r \\ \gamma \frac{v}{c} k'_r + \gamma k'_r \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \gamma k' - \gamma \frac{v}{c} k' \\ -\gamma \frac{v}{c} k' + \gamma k' \\ 0 \\ 0 \end{pmatrix}. \quad (10)$$

Therefore we obtain

$$\omega_r = \gamma \left(1 - \frac{v}{c} \right) \omega' = \gamma^2 \left(1 - \frac{v}{c} \right)^2 \omega = \frac{c-v}{c+v} \omega < \omega. \quad (11)$$

The reflected wave is therefore *redshifted*.

Problem 2 Lorentz Transformation of the Electromagnetic Fields

Consider an infinitely long cylinder with radius R , a homogeneous charge density $\rho = \alpha$ inside the cylinder, and vanishing current density $\mathbf{J} = 0$.

(2.a) Determine $\mathbf{E}$ and $\mathbf{B}$ outside the cylinder.

(2.b) An observer O' moves with velocity v parallel to the axis of the cylinder. What are the charge density ρ' and the current density $\mathbf{J}'$ measured by O' ?

(2.c) Use the methods of electrostatics and magnetostatics to calculate the fields $\mathbf{E}'$ and $\mathbf{B}'$ in the reference frame of O' .

(2.d) Show that the results from (2.c) agree with the Lorentz transformation of the results from (2.a).

Solution:

- (2.a) We work in cylindrical coordinates and choose the coordinate system such that the cylinder is parallel to the z -axis. Due to rotational symmetry about z and translation invariance along z , we know that the electric field can only have radial components

$$\mathbf{E}(\mathbf{r}) = E_\rho \hat{\rho}. \quad (12)$$

Then we integrate Gauss's law over a cylinder V around the z -axis with radius $r > R$ and height L , and obtain

$$\int_V \nabla \cdot \mathbf{E} = \frac{1}{\epsilon_0} \int_V \rho(r) \quad (13)$$

$$\Rightarrow \int_{\partial V} \mathbf{E} \cdot d\mathbf{S} = \frac{\alpha \pi R^2 L}{\epsilon_0} \quad (14)$$

$$\Rightarrow 2\pi r L E_r = \frac{\alpha \pi R^2 L}{\epsilon_0} \quad (15)$$

$$\Rightarrow E_r = \frac{\alpha R^2}{2\epsilon_0 r}. \quad (16)$$

Since there are no currents, $\mathbf{B} = 0$.

- (2.b) The four-current in the rest frame of the cylinder is

$$J^\mu = (c\rho, \mathbf{J}) = (c\alpha, 0, 0, 0). \quad (17)$$

In the reference frame of O' we obtain, by a Lorentz transformation with $\beta = v/c$,

$$J^\mu \rightarrow J'^\mu = \Gamma^\mu{}_\nu J^\nu = (\gamma c\alpha, 0, 0, -\gamma \beta c\alpha). \quad (18)$$

Thus we find

$$\rho' = \gamma \alpha, \quad (19)$$

$$\mathbf{J}' = -\gamma \alpha v \hat{\mathbf{z}}. \quad (20)$$

- (2.c) For the densities ρ' and $\mathbf{J}'$, in analogy to (2.a) with $R' = R$, we find

$$\mathbf{E}' = \frac{\gamma \alpha R^2}{2\epsilon_0 r} \hat{\rho} \quad (21)$$

We determine the magnetic field using Ampère's law. For this, we take a circle of radius $r > R$ around the cylinder

$$\oint \mathbf{B}' \cdot d\mathbf{l} = 2\pi r B'_\varphi = \mu_0 \pi R^2 J'_z \quad (22)$$

$$\Rightarrow B'_\varphi(r) = -\frac{\mu_0 \gamma \alpha v R^2}{2r} \quad (23)$$

Thus

$$\mathbf{B}'_\varphi(r) = -\frac{\mu_0 \gamma \alpha v R^2}{2r} \hat{\varphi}. \quad (24)$$

(2.d) Under a Lorentz boost along z , we find

$$\mathbf{E}'_{\perp} = \gamma (\mathbf{E}_{\perp} + \mathbf{v} \times \mathbf{B}) \quad (25)$$

$$= \gamma \mathbf{E}, \quad (26)$$

since $\mathbf{B} = 0$.

For the magnetic field $\mathbf{B}'$ we find

$$\mathbf{B}'_{\perp} = \gamma \left(\mathbf{B}_{\perp} - \frac{1}{c^2} \mathbf{v} \times \mathbf{E} \right) \quad (27)$$

$$= -\frac{\gamma v}{c^2} E_{\rho} \hat{\phi} \quad (28)$$

$$= -\frac{\gamma v \alpha R^2}{2\epsilon_0 r c^2} \hat{\phi} \quad (29)$$

$$= -\frac{\mu_0 \gamma v \alpha R^2}{2r} \hat{\phi}. \quad (30)$$

Thus both methods yield the same result.

Problem 3 (central exercise) Cherenkov Radiation

Consider a particle with charge q moving on the *spacelike* worldline

$$X(t) = Y(t) = 0, \quad Z(t) = vt \quad \text{with} \quad v > c. \quad (31)$$

Of course, such motion with $v > c$ is unphysical, but there is no mathematical inconsistency in using such a source term in Maxwell's equations. In media with $n > 1$, charged particles can even move with $v > c/n$; see subproblem (e).

Assume an observer is at rest at $\mathbf{r}_0 = (x_0, 0, 0)$ with $x_0 > 0$.

(3.a) Show that for

$$t < t_0 \equiv \frac{x_0}{c} \sqrt{1 - \left(\frac{c}{v}\right)^2}, \quad (32)$$

the past light cone of the observer at $(t, \mathbf{r}_0)$ does not intersect the worldline of the point charge.

This means that the retarded solution at the observer's position vanishes for $t < t_0$.

(3.b) Show that for $t > t_0$ the past light cone of the observer at $(t, \mathbf{r}_0)$ intersects the worldline of the charged particle at two events.

(3.c) Give the retarded solution for the potentials $\phi, \mathbf{A}$ at the point $(t, \mathbf{r}_0)$ for $t > t_0$. **Hint:** Work analogously to the derivation of the Liénard-Wiechart potential for charged particles moving on timelike worldlines.

(3.d) Show that the solution from (c) becomes singular at the time $t = t_0$.

(3.e) Determine the angle $\cos\theta_C$ with respect to the z -axis at which the particle would have to emit light so that the observer receives the signal at the time t_0 .

- (3.f) Generalize the results for t_0 and θ_C for a particle moving with velocity $v > c/n$ in a medium with refractive index n . **Remark:** This case is physically possible and is also frequently observed. This phenomenon is called *Cherenkov radiation*.

Solution:

- (3.a) The worldline of the charged particle $X^\mu = (t', 0, 0, vt')$ intersects the past light cone of the observer at the point $x_B^\mu = (t, \mathbf{r}_0)$ if and only if

$$(x_B^\mu - X^\mu)(x_{B,\mu} - X_\mu) = 0 \quad (33)$$

$$\Rightarrow -c^2(t - t')^2 + x_0^2 + (vt')^2 = 0 \quad (34)$$

$$\Rightarrow (v^2 - c^2)t'^2 + 2c^2 t t' + x_0^2 - c^2 t^2 = 0. \quad (35)$$

This quadratic equation has a solution only if

$$\Delta = (2c^2 t)^2 - 4(v^2 - c^2)(x_0^2 - c^2 t^2) > 0 \quad (36)$$

$$\Rightarrow t^2 > \frac{x_0^2(v^2 - c^2)}{(vc)^2} \quad (37)$$

$$\Rightarrow t > \frac{x_0}{c} \sqrt{1 - \left(\frac{c}{v}\right)^2} \quad (38)$$

- (3.b) For $t > t_0$, $\Delta > 0$. Thus we find two solutions for the intersection

$$t_\pm^* = \frac{-c^2 t \pm \sqrt{(vct)^2 - (v^2 - c^2)x_0^2}}{(v^2 - c^2)}. \quad (39)$$

- (3.c) We use the retarded Green's function and obtain the general expression

$$\phi(t, \mathbf{r}) = \frac{q}{4\pi\epsilon_0} \int \frac{\delta(\mathbf{r}' - \mathbf{X}(t_{\text{ret}}))}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}', \quad (40)$$

$$\mathbf{A}(t, \mathbf{r}) = \frac{q\mu_0}{4\pi} \int \frac{d\mathbf{X}}{dt} \bigg|_{t=t_{\text{ret}}} \frac{\delta(\mathbf{r}' - \mathbf{X}(t_{\text{ret}}))}{|\mathbf{r} - \mathbf{r}'|} d^3\mathbf{r}', \quad (41)$$

with $t_{\text{ret}} = t - \frac{|\mathbf{r} - \mathbf{r}'|}{c}$.

To evaluate the delta distribution, we use the two zeros/intersection points that we calculated above

$$\mathbf{r}_\pm^* = (0, 0, vt_\pm^*). \quad (42)$$

Thus

$$\delta(\mathbf{r}' - \mathbf{X}(t_{\text{ret}})) = \sum_{\mathbf{r}^* = \mathbf{r}_\pm^*} \frac{\delta(\mathbf{r}' - \mathbf{r}^*)}{|\det J(\mathbf{r}^*)|}, \quad (43)$$

Here J_{ij} is the Jacobian $\frac{\partial(\mathbf{r}' - \mathbf{X}(t_{\text{ret}}))_i}{\partial r_j'}$. We find

$$|\det J(\mathbf{r}^*)| = 1 - \frac{\mathbf{r} - \mathbf{r}^*}{c|\mathbf{r} - \mathbf{r}^*|} \frac{d\mathbf{X}}{dt}(t_{\text{ret}}) = 1 - \frac{v}{c} \frac{z - z^*}{|\mathbf{r} - \mathbf{r}^*|}. \quad (44)$$

Now considering the point $(t, \mathbf{x}_0) = (t, x_0, 0, 0)$, the expression simplifies to

$$\phi(t, \mathbf{x}_0) = \frac{q}{4\pi\epsilon_0} \sum_{z^*=z_{\pm}^*} \frac{1}{|\mathbf{r} - \mathbf{r}'|} \left| \frac{1}{1 - \frac{v}{c} \frac{-z^*}{|\mathbf{r} - \mathbf{r}^*|}} \right| \quad (45)$$

$$= \frac{q}{4\pi\epsilon_0} \sum_{z^*=z_{\pm}^*} \frac{1}{\left| |\mathbf{r} - \mathbf{r}^*| + \frac{v}{c} z^* \right|} \quad (46)$$

$$= \frac{q}{4\pi\epsilon_0} \sum_{z^*=z_{\pm}^*} \frac{1}{\left| \sqrt{x_0^2 + (z^*)^2} + \frac{v}{c} z^* \right|}. \quad (47)$$

For the vector potential we obtain analogously

$$\mathbf{A}(t, \mathbf{x}_0) = \frac{q\mu_0 v}{4\pi} \sum_{z^*=z_{\pm}^*} \frac{\hat{\mathbf{z}}}{\left| \sqrt{x_0^2 + (z^*)^2} + \frac{v}{c} z^* \right|}. \quad (48)$$

(3.d) At the time $t \rightarrow t_0$ we have

$$z_{\pm}^* = -\frac{c^2 t_0}{v^2 - c^2} = -\frac{-cx_0}{\sqrt{v^2 - c^2}}. \quad (49)$$

Substituting this into the expression above, we obtain

$$\phi(t, \mathbf{x}_0) = \frac{q}{2\pi\epsilon_0} \frac{1}{\left| \sqrt{x_0^2 + (z^*)^2} + \frac{v}{c} z^* \right|} \quad (50)$$

$$\rightarrow \frac{q}{2\pi\epsilon_0} \frac{1}{\left| \frac{cx_0}{\sqrt{v^2 - c^2}} - \frac{cx_0}{\sqrt{v^2 - c^2}} \right|} \quad (51)$$

$$\rightarrow \frac{q}{2\pi\epsilon_0} \frac{1}{0}. \quad (52)$$

The vector potential diverges in exactly the same way.

(3.e) We have already shown that the light cone of the observer at $(t = t_0, \mathbf{r}_0)$ intersects the worldline of the charged particle only at the point

$$\left(\underbrace{\frac{-cx_0}{v\sqrt{v^2 - c^2}}}_{t_0^*}, 0, 0, \underbrace{\frac{-cx_0}{\sqrt{v^2 - c^2}}}_{z_0} \right), \quad (53)$$

The angle between the signal connecting the observer at the point $\mathbf{r}_0 = (x_0, 0, 0)$ to the particle at the point $(0, 0, z_0)$ and the z -axis is then given by

$$\cos\theta_C = \frac{z_0}{\sqrt{z_0^2 + x_0^2}} = \frac{c}{v}. \quad (54)$$

(3.f) In a linear, isotropic medium with refractive index n , one can simply replace the speed of light c by c/n .

$$t_0^{(n)} = \frac{nx_0}{c} \sqrt{1 - \left(\frac{c}{nv}\right)^2}, \quad (55)$$

$$\cos\theta_C^{(n)} = \frac{c}{vn}. \quad (56)$$